MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DIODE, PAIRS, DIGITAL, KEY CODE KDF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all departments and Agencies of the department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the procurement of an electronic diode module.

1.2 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Name	Diode, pairs
Family	Digital
Key code	KDF
Size	1A
Weight	45 grams maximum
Failure rate	0.82 failures/10^6 hours maximum
Environment	Class I as specified in MIL-STD-1389
Maximum power dissipation	1.25 W
Forward voltage drop	
I_F = 0.5 mA	0.31 V minimum, 0.68 V maximum
I_F = 3.0 mA	0.39 V minimum, 0.77 V maximum
I_F = 10 mA	0.45 V minimum, 0.82 V maximum
I_F = 100 mA	0.56 V minimum, 0.93 V maximum
I_F = 250 mA	0.60 V minimum, 1.05 V maximum
Reverse current	
V_R = 10 V	0.2 µA maximum
V_R = 400 V	3.0 µA maximum
Continuous forward diode current	250 mA maximum
Peak one cycle surge current (60 Hz)	5 A maximum

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commander, Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, DC, 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5963

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1.3 Module function. This module contains 12 separate diode pairs with each pair connected in series. Each diode is externally accessible.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specifiction to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.82 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be KDF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 45 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/49-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

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3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with notes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. The following test equipment is necessary to conduct acceptance tests for this module. Test equipment shall be as specified in MIL-STD-1665 and that listed below

Equipment	MIL-STD-1665 designator	No. required (maximum per test figure)
DC power supply	Item P01	1
DC voltmeter	Item V04	1
Megohm bridge	Item R06	1
Ohmmeter	Item R05	1
DC power supply	Item P03	1
DC ammeter	Item X03	1
Oscilloscope	Item D02	1
Resistors		
86.6Ω ±1%, 1/2 W		1
100 kΩ ±1%, 1/4 W		1
68.1 kΩ ±1%, 1/4 W		1
11.0 kΩ ±1%, 1/4 W		1
3.32 kΩ ±1%, 1/2 W		1
332Ω ±1%, 4 W		1
133Ω ±1%, 10 W		1

All resistors shall be metal film unless otherwise specified.

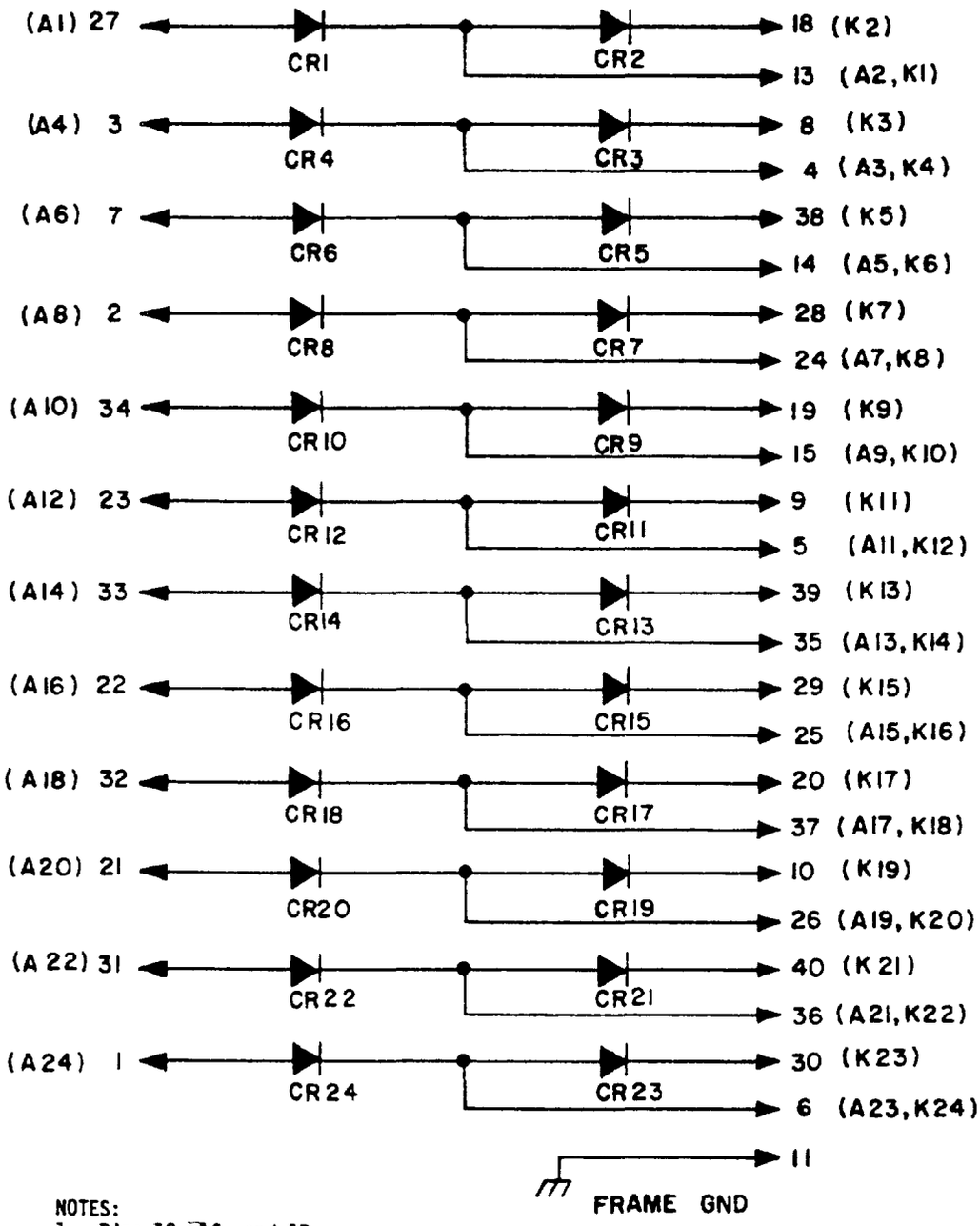
4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through L shall be performed in any order with the following exceptions. If test conditions L, A, B, or J are part of a sequence of tests, the order of testing shall be as follows L, A, B, J where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be perfomed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

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NOTES:
1. Pins 12, 16, and 17 are not used and each shall be isolated from all other pins.
2 A = Anode, K = Cathode.

FIGURE 1 KDF module function diagram,diode, pairs.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	II	X	10	-	1	-	10	-	1	-	MΩ
B1-B4	Integrity	II	S	10	-	1	-	10	-	1	-	MΩ
C	Frame continuity	III	S	0	1	0	1	0	1	0	1	Ω
D1-D24	Forward voltage drop (I_F = 0.5 mA)	IV	X	0.43	0.62	0.41	0.64	0.35	0.66	0.31	0.68	V
E1-E24	Forward voltage drop (I_F = 3 mA)	IV	S	0.52	0.69	0.50	0.71	0.43	0.75	0.39	0.77	V
F1-F24	Forward voltage drop (I_F = 10 mA)	IV	X	0.57	0.75	0.55	0.77	0.49	0.80	0.45	0.82	V
G1-G24	Forward voltage drop (I_F = 100 mA)	IV	S	0.68	0.87	0.66	0.89	0.60	0.91	0.56	0.93	V
H1-H24	Forward voltage drop (I_F = 250 mA)	IV	X	0.71	0.97	0.69	0.99	0.64	1.02	0.60	1.05	V
J1-J24	Reverse current (V_R = 400 V)	V	X	0	0.15	0	0.2	0	1.5	0	3.0	µA
K1-K24	Reverse current (V_R = 10 V)	V	S	0	0.01	0	0.02	0	0.1	0	0.2	µA
L1-L24	Peak one cycle surge current	VI	Q	5.0	--	--	--	--	--	--	--	A

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through L in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through K shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.1.2 Critical component temperature test. The critical component temperature (CCT) tests shall be conducted in accordance with MIL-M-28787. During the CCT tests, the module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall not be operated. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A through K shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through K shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/49-1 supersedes part number M28787/49. The two parts are interchangeable.

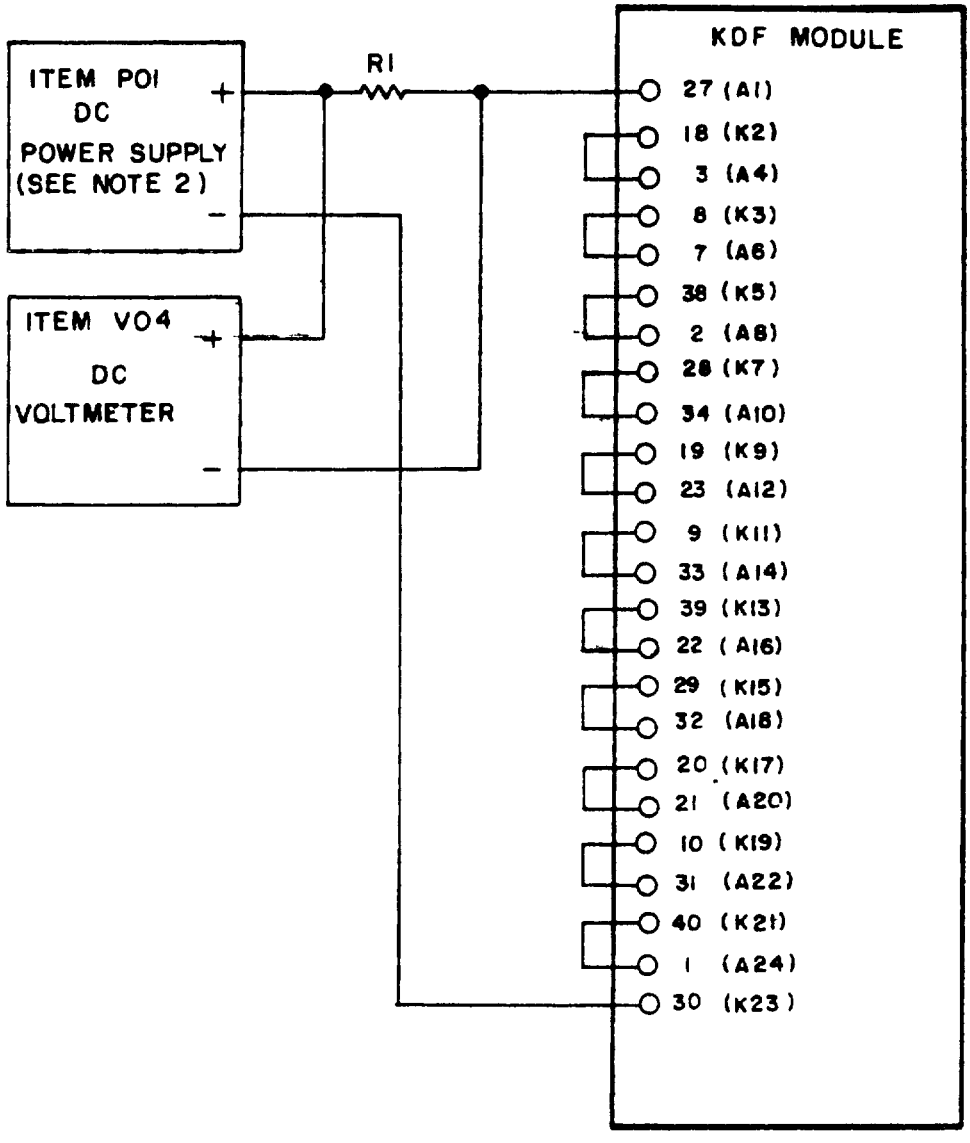
6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity
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NOTES:
1. R1 = 86.6Ω ±1%, 1/2 W.
2. Adjust item PO1 for a voltage reading of 5.00 V ±1% on item VO4, corresponding to a current of 57.7 mA.

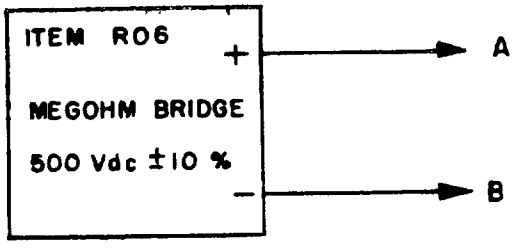
FIGURE 2. Operating temperature, component temperature, critical component temperature, and life test circuit.

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TABLE II. Isolation and integrity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	1-10,12-40	11	10	∞	MΩ	3
B1	Integrity	2,7,12-15, 17-20,22,24, 25,27,28,29, 32,34,37,38	1,3-6,8-11,16, 21,23,26,30, 31,33,35,36, 39,40	↓	↓	↓	↓
B2	↓	3,4,5,7,8,9, 13-19,23,27, 33,34,35,38, 39	1,2,6,10,11, 12,20,21,22, 24,25,26, 28-32,36,37, 40	↓	↓	↓	↓
B3	↓	1-4,6,7,8,14, 15,19,20,24, 28,30-40	5,9-13,16,17, 18,21,22,23, 25,26,27,29	↓	↓	↓	↓
B4	Integrity	1-10,13,14, 18,21-30,38	11,12,15,16, 17,19,20, 31-37,39,40	10	-	MΩ	3

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WARNING

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

KDF MODULE	
27 (A1)	(K2) 18
	(A2,K1) 13
3 (A4)	(K3) 8
	(A3,K4) 4
7 (A6)	(K5) 38
	(A5,K6) 14
2 (A8)	(K7) 28
	(A7,K8) 24
34(A10)	(K9) 19
	(A9,K10) 15
23 (A12)	(K11) 9
	(A11,K12) 5
33 (A14)	(K13) 39
	(A13,K14) 35
22 (A16)	(K15) 29
	(A15,K16) 25
32 (A18)	(K17) 20
	(A17,K18) 37
21 (A20)	(K19) 10
	(A19,K20) 26
31 (A22)	(K21) 40
	(A21,K22) 36
1 (A24)	(K23) 30
	(A23,K24) 6
11 (FRAME GND)	

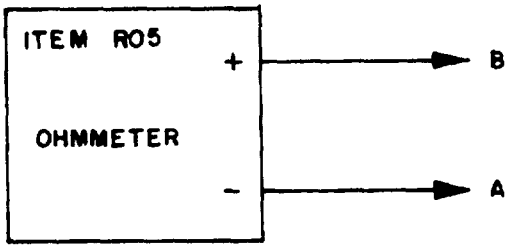
FIGURE 3. Isolation and integrity test circuits for table II.

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TABLE III. Frame continuity test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C	Frame continuity	11	Module fin	0	1	Ω	4

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NOTE Penetrate the anodized finish of the module fin with a sharply pointed probe and measure the resistance between the fin and the pin.

KDF MODULE	
27 (A1)	(K2) 18 (A2,K1) 13
3 (A4)	(K3) 8 (A3,K4) 4
7 (A6)	(K5) 38 (A5,K6) 14
2 (A8)	(K7) 28 (A7,K8) 24
34(A10)	(K9) 19 (A9,K10) 15
23 (A12)	(K11) 9 (A11,K12) 5
33 (A14)	(K13) 39 (A13,K14) 35
22 (A16)	(K15) 29 (A15,K16) 25
32 (A18)	(K17) 20 (A17,K18) 37
21 (A20)	(K19) 10 (A19,K20) 26
31 (A22)	(K21) 40 (A21,K22) 36
1 (A24)	(K23) 30 (A23,K24) 6
11 (FRAME GND)	

FIGURE 4. Frame continuity test for table III.

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TABLE IV. Forward voltage drop test.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A and G to pin	F to pin	Min	Max		
D1	V_F (I_F = 0.5 mA)	27	13	0.43	0.62	V	5
D2		13	18				
D3		4	8				
D4		3	4				
D5		14	38				
D6		7	14				
D7		24	28				
D8		2	24				
D9		15	19				
D10		34	15				
D11		5	9				
D12		23	5				
D13		35	39				
D14		33	35				
D15		25	29				
D16		22	25				
D17		37	20				
D18	V_F (I_F = 0.5 mA)	32	37	0.43	0.62	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		A and G to pin	F to pin	Min	Max		
D19	V_F (I_F = 0.5 mA)	26	10	0.43	0.62	V	5
D20	↓	21	26	↓	↓	↓	↓
D21		36	40				
D22		31	36				
D23		6	30				
D24	V_F (I_F = 0.5 mA)	1	6	0.43	0.62	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		B and G to pin	F to pin	Min	Max		
E1	V_F (I_F = 3.0 mA)	27	13	0.52	0.69	V	5
E2		13	18				
E3		4	8				
E4		3	4				
E5		14	38				
E6		7	14				
E7		24	28				
E8		2	24				
E9		15	19				
E10		34	15				
E11		5	9				
E12		23	5				
E13		35	39				
E14		33	35				
E15		25	29				
E16		22	25				
E17		37	20				
E18	V_F (I_F = 3.0 mA)	32	37	0.52	0.69	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° + 5°C Ambient		Unit	Test figure
		B and G to pin	F to pin	Min	Max		
E19	V_F (I_F = 3.0 mA)	26	10	0.52	0.69	V	5
E20	↓	21	26	↓	↓	↓	↓
E21		36	40				
E22		31	36				
E23		6	30				
E24	V_F (I_F = 3.0 mA)	1	6	0.52	0.69	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		C and G to pin	F to pin	Min	Max		
F1	V_F (I_F = 10 mA)	27	13	0.57	0.75	V	5
F2		13	18				
F3		4	8				
F4		3	4				
F5		14	38				
F6		7	14				
F7		24	28				
F8		2	24				
F9		15	19				
F10		34	15				
F11		5	9				
F12		23	5				
F13		35	39				
F14		33	35				
F15		25	29				
F16		22	25				
F17		37	20				
F18	V_F (I_F = 10 mA)	32	37	0.57	0.75	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		C and G to pin	F to pin	Min	Max		
F19	V_F (I_F = 10 mA)	26	10	0.57	0.75	V	5
F20	↓	21	26	↓	↓	↓	↓
F21		36	40				
F22		31	36				
F23		6	30				
F24	V_F (I_F = 10 mA)	1	6	0.57	0.75	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		D and G to pin	F to pin	Min	Max		
G1	V_F (I_F = 100 mA)	27	13	0.68	0.87	V	5
G2		13	18				
G3		4	8				
G4		3	4				
G5		14	38				
G6		7	14				
G7		24	28				
G8		2	24				
G9		15	19				
G10		34	15				
G11		5	9				
G12		23	5				
G13		35	39				
G14		33	35				
G15		25	29				
G16		22	25				
G17		37	20				
G18	V_F (I_F = 100 mA)	32	37	0.68	0.87	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		D and G to pin	F to pin	Min	Max		
G19	V_F (I_F = 100 mA)	26	10	0.68	0.87	V	5
G20	↓	21	26	↓	↓	↓	↓
G21		36	40				
G22		31	36				
G23		6	30				
G24	V_F (I_F = 100 mA)	1	6	0.68	0.87	V	5

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TABLE IV. Forward voltage drop test - Continued.

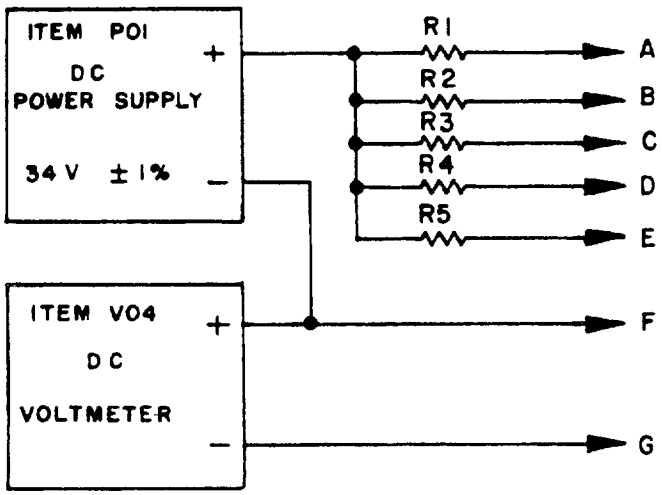
Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		E and G to pin	F to pin	Min	Max		
H1	V_F (I_F = 250 mA)	27	13	0.71	0.97	V	5
H2		13	18				
H3		4	8				
H4		3	4				
H5		14	38				
H6		7	14				
H7		24	28				
H8		2	24				
H9		15	19				
H10		34	15				
H11		5	9				
H12		23	5				
H13		35	39				
H14		33	35				
H15		25	29				
H16		22	25				
H17		37	20				
H18	V_F (I_F = 250 mA)	32	37	0.71	0.97	V	5

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TABLE IV. Forward voltage drop test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ± 5°C Ambient		Unit	Test figure
		E and G to pin	F to pin	Min	Max		
H19	V_F (I_F = 250 mA)	26	10	0.71	0.97	V	5
H20	↓	21	26	↓	↓	↓	↓
H21		36	40				
H22		31	36				
H23		6	30				
H24	V_F (I_F = 250 mA)	1	6	0.71	0.97	V	5

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KDF MODULE	
27 (A1)	(K2) 18
	(A2,K1) 13
3 (A4)	(K3) 8
	(A3,K4) 4
7 (A6)	(K5) 38
	(A5,K6) 14
2 (A8)	(K7) 28
	(A7,K8) 24
34 (A10)	(K9) 19
	(A9,K10) 15
23 (A12)	(K11) 9
	(A11,K12) 5
33 (A14)	(K13) 39
	(A13,K14) 35
22 (A16)	(K15) 29
	(A15,K16) 25
32 (A18)	(K17) 20
	(A17,K18) 37
21 (A20)	(K19) 10
	(A19,K20) 26
31 (A22)	(K21) 40
	(A21,K22) 36
1 (A24)	(K23) 30
	(A23,K24) 6
11 (FRAME GND)	

NOTES.
1. R1 = 68.1 kΩ ±1%, 1/4 W.
2. R2 = 11.0 kΩ ±1%, 1/4 W.
3. R3 = 3.32 kΩ ±1%, 1/2 W
4. R4 = 332Ω ±1%, 4 W
5. R5 = 133Ω ±1%, 10 W

FIGURE 5 Forward voltage drop test circuit for table IV

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TABLE V. Reverse current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Reverse current (V_R = 400 V)	13	27	0	0.15	µA	6
J2		18	13				
J3		8	4				
J4		4	3				
J5		38	14				
J6		14	7				
J7		28	24				
J8		24	2				
J9		19	15				
J10		15	34				
J11		9	5				
J12		5	23				
J13		39	35				
J14		35	33				
J15		29	25				
J16		25	22				
J17		20	37				
J18	Reverse current (V_R = 400 V)	37	32	0	0.15	µA	6

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TABLE V. Reverse current test - Continued.

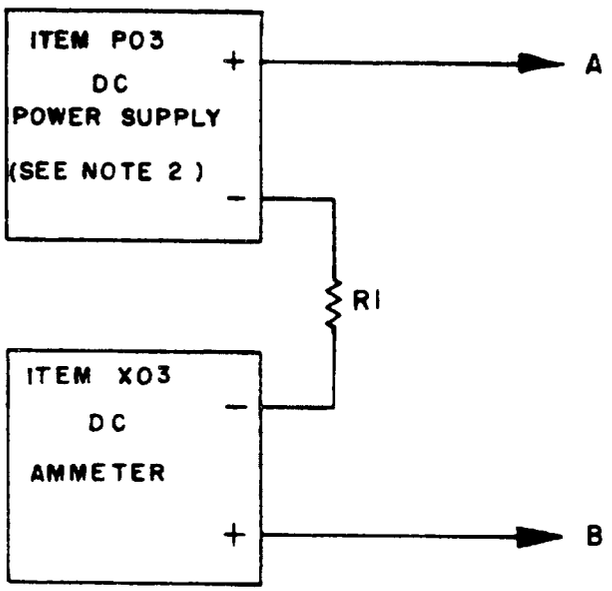
Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J19	Reverse current (V_R = 400 V)	10	26	0	0.15	µA	6
J20		26	21				
J21		40	36				
J22		36	31				
J23		30	6				
J24	Reverse current (V_R = 400 V)	6	1		0.15		
K1	Reverse current (V_R = 10 V)	13	27		0.01		
K2		18	13				
K3		8	4				
K4		4	3				
K5		38	14				
K6		14	7				
K7		28	24				
K8		24	2				
K9		19	15				
K10		15	34				
K11	Reverse current (V_R = 10 V)	9	5	0	0.01	µA	6

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TABLE V. Reverse current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K12	Reverse current (V_R = 10 V)	5	23	0	0.01	µA	6
K13		39	35				
K14		35	33				
K15		29	25				
K16		25	22				
K17		20	37				
K18		37	32				
K19		10	26				
K20		26	21				
K21		40	36				
K22		36	31				
K23		30	6				
K24	Reverse current (V_R = 10 V)	6	1	0	0.01	µA	6

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WARNING

High voltage (400 Vdc) will be present at the output binding posts of item P03 during test condition J. Always reduce the output voltage to zero volts before connecting and disconnecting the leads.

CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

NOTES:
1. R1 = 100 kΩ ±1%, 1/4 W.
2. For test conditions J1-J24 set item P03 at 400 V. For test conditions K1-K24 set item P03 at 10 V.

KDF MODULE	
27 (A1)	(K2) 18
	(A2,K1) 13
3 (A4)	(K3) 8
	(A3,K4) 4
7 (A6)	(K5) 38
	(A5,K6) 14
2 (A8)	(K7) 28
	(A7,K8) 24
34(A10)	(K9) 19
	(A9,K10) 15
23 (A12)	(K11) 9
	(A11,K12) 5
33 (A14)	(K13) 39
	(A13,K14) 35
22 (A16)	(K15) 29
	(A15,K16) 25
32 (A18)	(K17) 20
	(A17,K18) 37
21 (A20)	(K19) 10
	(A19,K20) 26
31 (A22)	(K21) 40
	(A21,K22) 36
1 (A24)	(K23) 30
	(A23,K24) 6
11 (FRAME GND)	

FIGURE 6. Reverse current test circuit for table V.

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TABLE VI. Peak one cycle surge current test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Peak one cycle surge current	27	13	5.0	-	A	7
L2		13	18				
L3		4	8				
L4		3	4				
L5		14	38				
L6		7	14				
L7		24	28				
L8		2	24				
L9		15	19				
L10		34	15				
L11		5	9				
L12		23	5				
L13		35	39				
L14		33	35				
L15		25	29				
L16		22	25				
L17		37	20				
L18	Peak one cycle surge current	32	37	5.0	-	A	7

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TABLE VI. Peak one cycle surge current test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L19	Peak one cycle surge current	26	10	5.0	-	A	7
L20	↓	21	26	↓	↓	↓	↓
L21		36	40				
L22		31	36				
L23		6	30				
L24	Peak one cycle surge current	1	6	5.0	-	A	7

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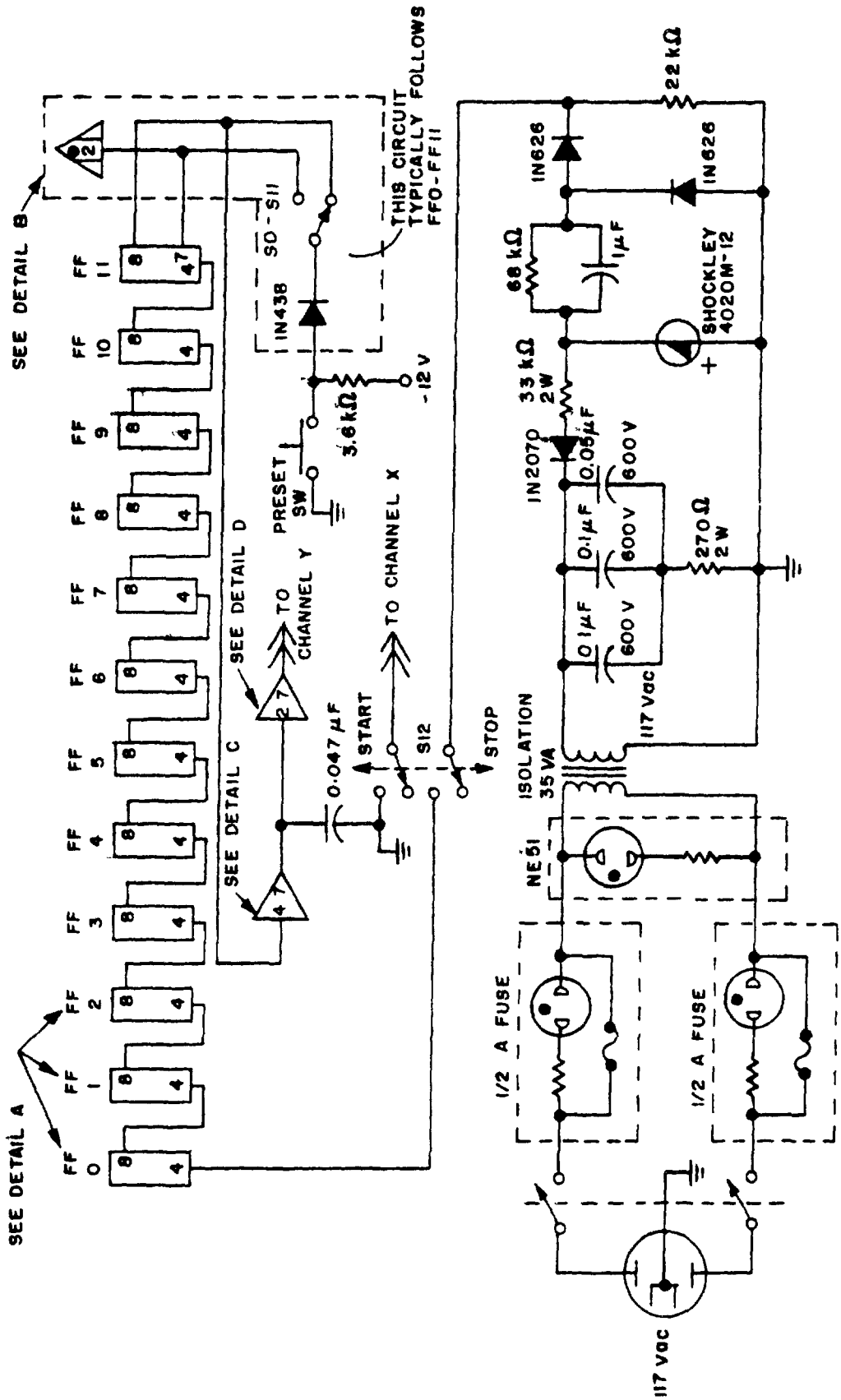


FIGURE 7. Peak one cycle surge current test circuit for table VI.

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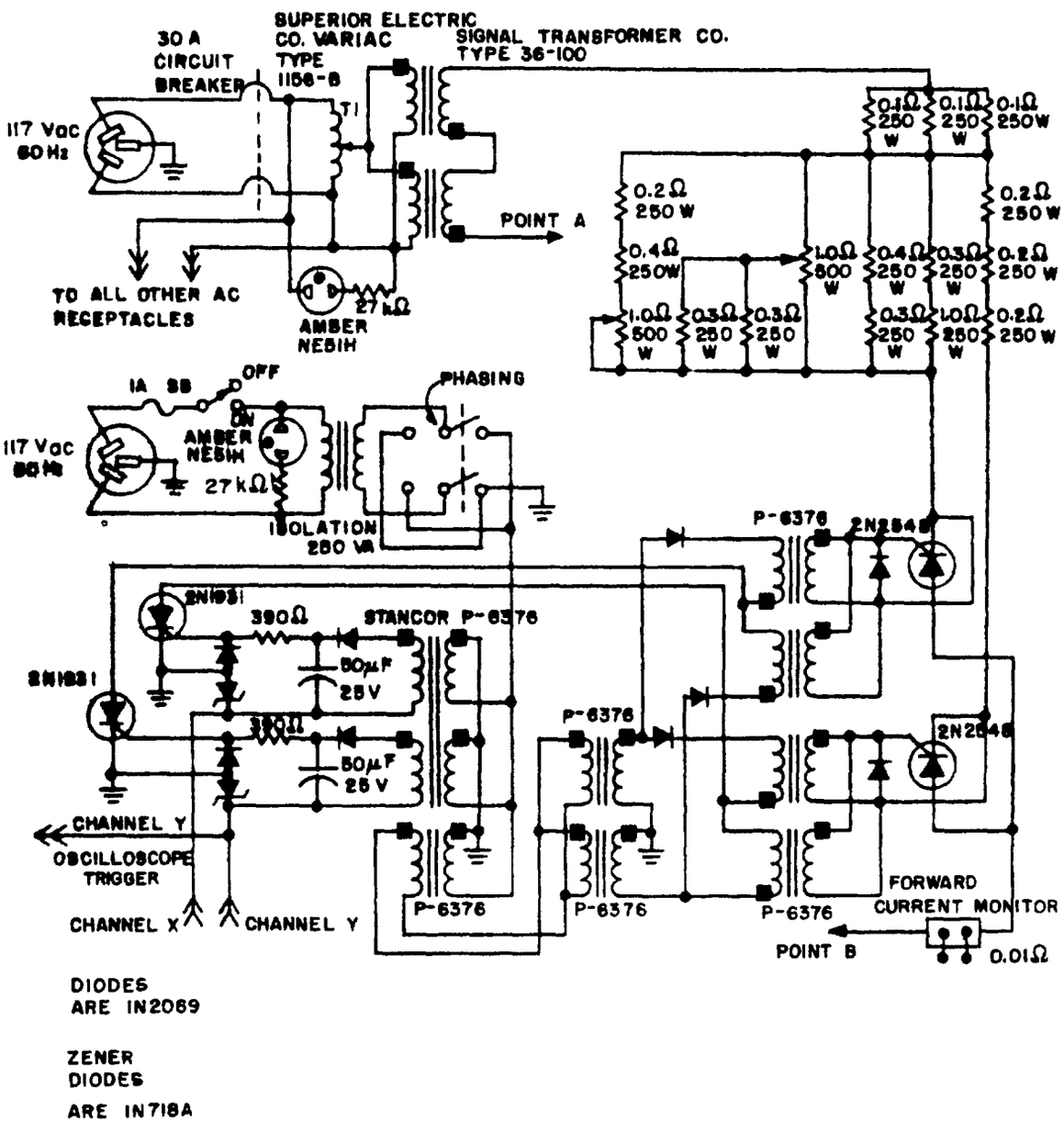


FIGURE 7. Peak one cycle surge current test circuit for table VI - Continued.

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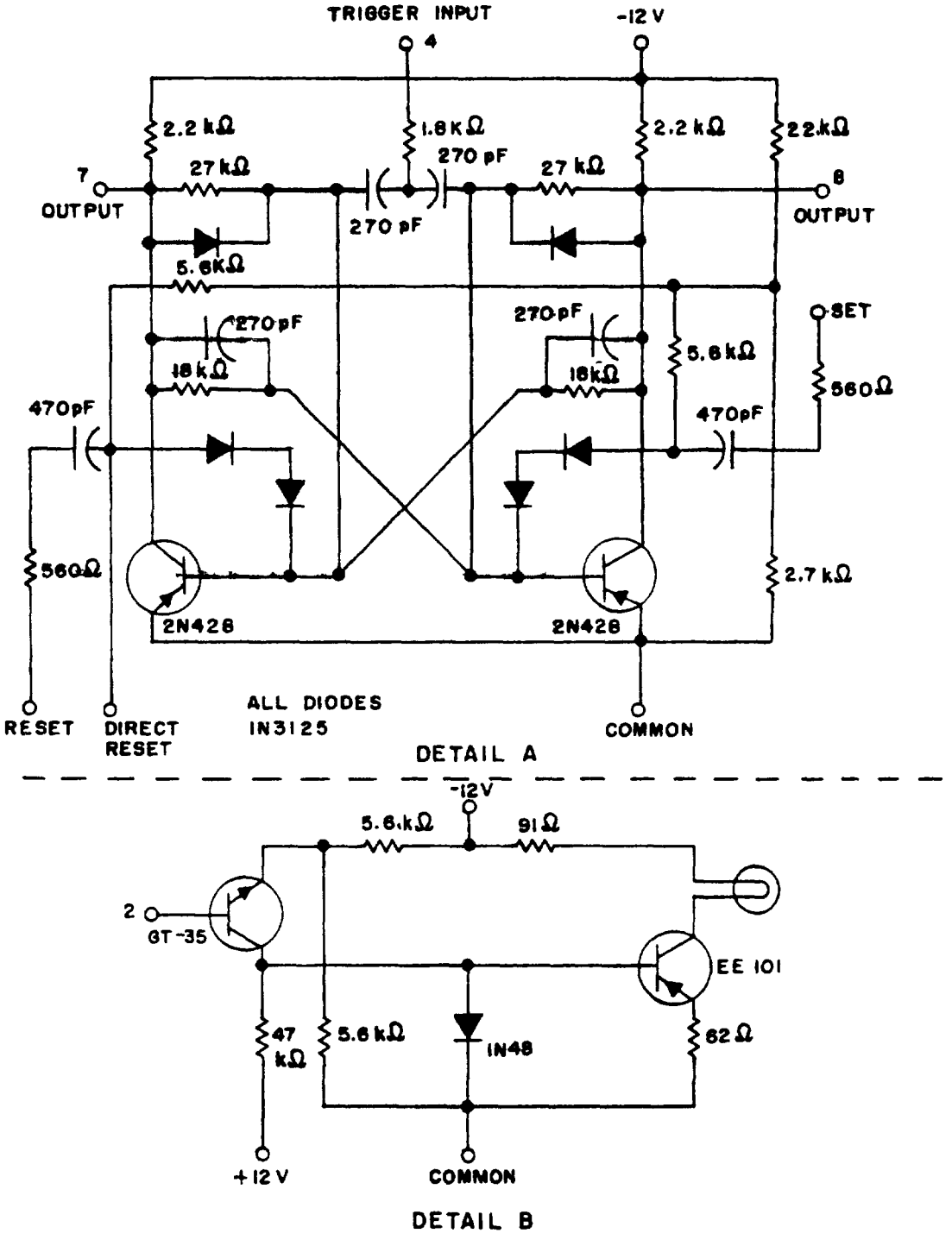


FIGURE 7. Peak one cycle surge current test circuit for table VI - Continued.

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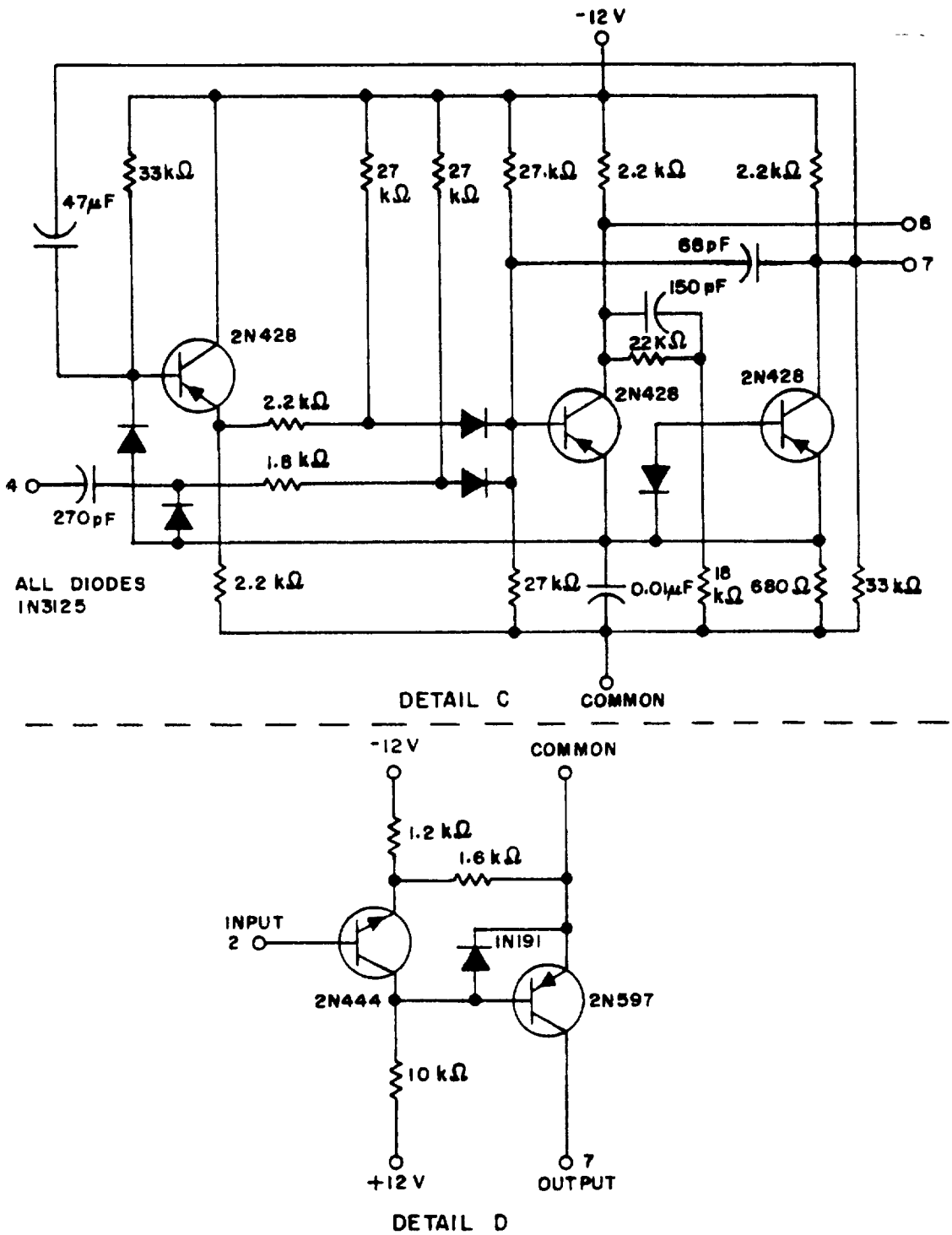
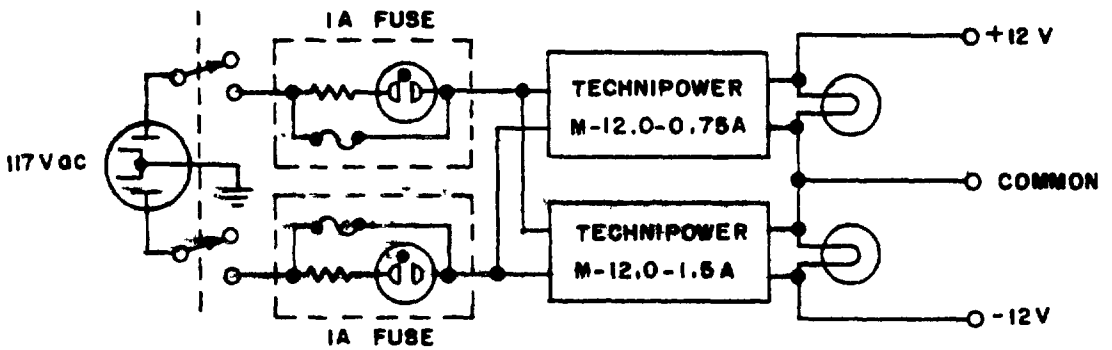


FIGURE 7. Peak one cycle surge current test circuit for table VI - Continued.

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NOTES
1 Switches S0 through S11 control the preset count of the binary counter. When the counter fills to all 1's, channel Y will allow one cycle of 60 Hz.
2. The binary counter shall be preset to a count of:

FF =	0	1	2	3	4	5	6	7	8	9	10	11
Output =	1	1	1	1	0	1	1	1	1	0	0	0

This count will result in 1 minute or 3,600 cycles of 60 Hz on channel X prior to the one cycle on channel Y.

3. Using a KDF set-up sample, with power switches "ON" and switch S12 in START, monitor current with item D02. Adjust variac T1 for a 7.86 mV indication on item D02 which corresponds to 786 A peak. A reading of .786 A peak equals an average current of 250 A (1/2 wave I avg = 318 I peak).
4. The 1.0Ω load potentiometers for channel Y should be adjusted for an oscilloscope indication of 50 mV which corresponds to a 5 A peak one cycle surge.
5. The peak one cycle surge current test shall be performed one time per test condition for each KDF module.
6 All resistors are 1/2 W, unless otherwise noted.

FIGURE 7. Peak one cycle surge current test circuit for table VI - Continued.

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